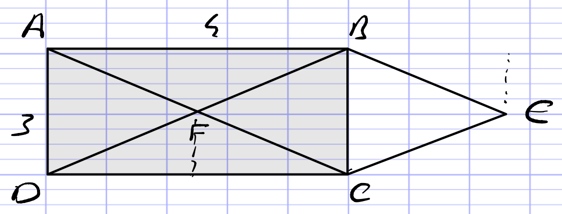
DS6 - Suites et P.S.

8'30"

⇒ Exercice n°1

4

1°/



1

2°/ a) $\vec{BA}\cdot\vec{BE} = -4\times 2 = -8$ 1

b) $\vec{CF}\cdot\vec{CD} = 4\times 2 = 8$ 1

c) $\vec{AF}\cdot\vec{AB} = 4\times 2 = 8$ 1

8'15"

⇒ Exercice n°2

5

1°/ $\vec{AB}\cdot\vec{AC} = AB\times AC\times\cos(\widehat{BAC})$ 1

$= 5\times 4\times\cos(45°)$

1.5/ $= 5\times 4\times\frac{\sqrt{2}}{2}$

$= 10\sqrt{2}$ 0.5

2°/ $\vec{AB}\cdot\vec{AC} = \vec{AH}\cdot\vec{AC}$ 0.5

$= AH\times AC$ 0.5 or $HC = \sqrt{5^2-4^2}$

2/ $= 4\times 7$ $= 3$ 0.5

$= 28$ 0.5 donc $AC = 4+3 = 7$

3°/ $\vec{AB}\begin{pmatrix}7\\1\end{pmatrix}$ 0.5 et $\vec{AC}\begin{pmatrix}4\\4\end{pmatrix}$ 0.5

1.5

Donc $\vec{AB}\cdot\vec{AC} = 7\times 4+1\times 4 = 28+4 = 32$ 0.5

2'00

2

=> Exercice n°3

$$\vec{OF}\cdot\vec{DC} = \vec{OF}\cdot(\vec{DO}+\vec{OC})$$ 0.5

$$= \vec{OF}\cdot\vec{DO} + \vec{OF}\cdot\vec{OC}$$ 0.5

$$= -\vec{OF}\cdot\vec{OD} + \vec{OF}\cdot\vec{OC}$$

$$= -OF\times OD\times\cos\left(\frac{\pi}{2}\right) + OF\times OC\times\cos\left(\frac{3\pi}{4}\right)$$

$$= 0 + 2\times2\times\frac{-\sqrt{2}}{2}$$ 0.5 0.5

$$= -2\sqrt{2}.$$

5'00"

6

12×0,5

=> Exercice n°4

1°/ (V_n) b

2°/ $U_0 = 0^2 = 0$ a c

3°/ $V_2 = 3\times V_1 = 3\times3\times V_0 = 3^2\times2 = 18$ c

4°/ $U_{n+1} = (n+1)^2 = n^2+2n+1$ b

5°/ nuage de point parabolique (car n^2) c

6°/ croissante a

7°/ $U_{n+1} - U_n = n^2+2n+1-n^2$
$= 2n+1$ c

8°/ $V_{n+1} = 3V_n$ donc b

9°/ $V_0 > 0$ et $q > 1$ donc a

10°/ $\forall n \in \mathbb{N}$, $V_n = V_0 \times q^n$
$= 2 \times 3^n$ (c)

11°/ $S = 1^{er} \times \dfrac{1 - q^{nbn}}{1-q} = 1 \times \dfrac{1-3^{11}}{1-3} = 88\,573$ (c)

3'00'' (4)

⇒ Exercice n°5

1 1°/ $\forall n \in \mathbb{N}$, $V_n = V_0 \times q^n = 12 \times \left(\frac{1}{4}\right)^n$ 0.25 0.25 0.25 0.25

1 2°/ $V_0 > 0$ 0.25 et $q < 1$ 0.25 donc (V_n) est décroissante 0.5

1 3°/ On cherche $N \in \mathbb{N}$ tq $V_N = \frac{3}{64}$ 0.25

$\Leftrightarrow 12 \times \left(\frac{1}{4}\right)^N = \frac{3}{64}$ 0.25

$\Leftrightarrow \left(\frac{1}{4}\right)^N = \frac{3}{64} \times \frac{1}{12}$

$\Leftrightarrow \left(\frac{1}{4}\right)^N = \frac{1}{256}$ 0.25

$\Leftrightarrow \frac{1}{4^N} = \frac{1}{4^4}$

$\Leftrightarrow N = 4$ 0.25

Donc $N = 4$

1 4°/ $S = 1^{er}\text{ terme} \times \dfrac{1 - \text{raison}^{nbn}}{1 - \text{raison}}$ 0.25

$= 12$ 0.25 $\times \dfrac{1 - \left(\frac{1}{4}\right)^{3}}{1 - \left(\frac{1}{4}\right)}$ 0.25

$= 16.$ 0.25